

	A	B	C	D	E	F
1	Physical Quantity	Formula / Relation	Dimensional Formula	C.G.S. Unit	M.K.S. / SI Unit	Symbol
2	Mass	-	[M]	g	kg	m
3	Distance / Displacement	-	[L]	cm	m	s
4	Time	-	[T]	s	s	t
5	Angle	$\Theta = \text{arc} / \text{radius}$	$[M^0 L^0 T^0]$	degree($^\circ$)	rad	-
6	Volume	-	$[L^3]$	cm^3	m^3	V
7	Density	$\rho = m/V$	$[ML^{-3}]$	g/cm^3	kg/m^3	ρ
8	Velocity	$v = \Delta s / \Delta t$	$[LT^{-1}]$	cm/s	m/s	v
9	Acceleration	$a = \Delta v / \Delta t$	$[LT^{-2}]$	cm/s^2	m/s^2	a
10	Momentum	$p = mv$	$[MLT^{-1}]$	$\text{g}\cdot\text{cm/s}$	$\text{kg}\cdot\text{m/s}$	p
11	Force	$F = ma$	$[MLT^{-2}]$	dyne	N	F
12	Energy / Work	$\Delta E = F\Delta s$	$[ML^2T^{-2}]$	erg	J	E
13	Kinetic Energy	$\frac{1}{2}mv^2$	$[ML^2T^{-2}]$	erg	J	KE
14	Gravitational Potential Energy	$E_p = mgh$	$[ML^2T^{-2}]$	erg	J	E_p
15	Energy Density	Energy / Volume	$[ML^{-1}T^{-2}]$	erg/cm^3	J/m^3	-
16	Power	$P = E/t$	$[ML^2T^{-3}]$	erg/s	W	P
17	Pressure	$P = F/A$	$[ML^{-1}T^{-2}]$	dyne/cm^2	N/m^2 or Pa	P
18	Time Period	-	[T]	s	s	T
19	Velocity Gradient	dv/dx	$[T^{-1}]$	s^{-1}	s^{-1}	-
20	Force Constant (Spring)	F/x	$[MT^{-2}]$	dyne/cm or g/s^2	N/m or kg/s^2	-
21	Stress	F/A	$[ML^{-1}T^{-2}]$	dyne/cm^2	N/m^2	-
22	Strain	$\Delta L / L$	$[M^0 L^0 T^0]$	-	-	-
23	Focal Length	-	[L]	cm	m	f
24	Refractive Index	$n = c / v$	-	-	No unit	n
25	Number of Bits (2^n)	2^n	-	-	No unit	n
26	Moment of Inertia	$I = Mk^2$	$[ML^2]$	$\text{g}\cdot\text{cm}^2$	$\text{kg}\cdot\text{m}^2$	I
27	Angular Momentum	$L = I\omega$	$[ML^2T^{-1}]$	$\text{g}\cdot\text{cm}^2/\text{s}$	$\text{kg}\cdot\text{m}^2/\text{s}$	L
28	Angular Displacement	$\Delta\theta = \Delta s / r$	$[M^0 L^0 T^0]$	radian or degree	radian or degree	-
29	Angular Velocity	$\omega = d\theta/dt$	$[T^{-1}]$	rad/s	rad/s	ω
30	Angular Acceleration	$\alpha = d\omega/dt$	$[T^{-2}]$	rad/s^2	rad/s^2	α
31	Critical Angle	-	-	degrees ($^\circ$)	degrees ($^\circ$)	c
32	Impulse	$F \times \text{time}$	$[MLT^{-1}]$	$\text{dyne}\cdot\text{s}$	$\text{N}\cdot\text{s}$	-
33	Torque	$F \times \text{distance}$	$[ML^2T^{-2}]$	$\text{dyne}\cdot\text{cm}$	$\text{N}\cdot\text{m}$	τ
34	Radius of Gyration	$k = \sqrt{I/m}$	[L]	cm	m	-
35	Frequency	$f = 1/T$	$[T^{-1}]$	cycle/s or Hz	Hz	f
36	Gravitational Constant	$G = Fd^2 / m_1 m_2$	$[M^{-1}L^3T^{-2}]$	$\text{dyne}\cdot\text{cm}^2/\text{g}^2$	$\text{N}\cdot\text{m}^2/\text{kg}^2$	G
37	Intensity of Gravitational Field	F/m	$[LT^{-2}]$	cm/s^2	m/s^2	-
38	Gravitational Potential	work/mass	$[L^2T^{-2}]$	cm^2/s^2	m^2/s^2	-
39	Modulus of Elasticity	stress/strain	$[ML^{-1}T^{-2}]$	dyne/cm^2	N/m^2	-
40	Charge	$Q = It$	[TI]	C	C	Q
41	Current	$I = \text{Voltage} / \text{Resistance}$	[I]	A	A	I
42	T-ratios (sin, cos, tan)	Length / Length	$[M^0 L^0 T^0]$	-	-	-
43	Surface Tension	force/length	$[ML^0T^{-2}]$	dyne/cm	N/m	γ
44	Poisson's Ratio	strain/strain	$[M^0 L^0 T^0]$	-	-	ν (NU)
45	Power (Optics)	$P = 1/f$	-	-	Dioptres (D)	P
46	Coefficient of Viscosity	$\eta = F\cdot dx / A\cdot dv$	$[ML^{-1}T^{-1}]$	poise or $\text{g}\cdot\text{cm}^{-1}\cdot\text{s}^{-1}$	$\text{Pa}\cdot\text{s}$ or $\text{kg}\cdot\text{m}^{-1}\cdot\text{s}^{-1}$	η